

Affine-product solutions and non-suspended mixed sweeps

Astrée Iriart

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Abstract

We classify polynomial solutions in an affine-product ansatz for the normalized mixed middle-branch equation $(G_y - \mathcal{L}_F F)^2 + 4F_y \mathcal{L}_F G = 0$, where $\mathcal{L}_F = \partial_x + 2F\partial_\theta$. Over a field of characteristic zero, for each integer $d \geq 1$, we take $F = (\theta + s(x))^d(y + a(x)) + b(x)$. Polynomial solvability forces $b = -s'/2$ and both a, s to have degree at most two. The remaining solutions are explicit cubic polynomials in $y + a(x)$, with one coefficient satisfying $9\mu^2 + 4(d-2)\mu + 4d^2 = 0$ and one free additive constant. The associated four-dimensional sweeps have geometric degree $2d+2$. A univariate generic-fiber polynomial proves this degree without a special-fiber count. Over the complex numbers the critical image has dimension three, whereas the triangular suspension class relevant to Gao's mixed-branch question has critical-image dimension at most two. Thus the family is not polynomially left–right equivalent to that suspension class, at every even geometric degree at least four. The classification is restricted to the stated ansatz; the sweeps themselves have nonconstant Jacobian determinant and are not Keller maps.

Contribution and significance

The contribution is an exact solution classification for one mixed-branch ansatz, together with its degree formula and a global non-suspension consequence. Gao's characteristic-invariance theorem [1, Theorem 4.6] identifies a locus on which solutions collapse to suspensions; our solutions lie outside that locus and have a different critical-image dimension. This gives an explicit family at every even geometric degree at least four, with the equivalence relation and the comparison class both specified.

The mechanism is elementary but restrictive: transverse-degree comparison forces a cubic, a nonresonant quadratic equation fixes its leading coefficient, and a lowest-order identity forces the compensating term $b = -s'/2$. The critical-image invariant itself is standard; the result is its explicit application to this classified family, not a general classification of polynomial maps. In particular, the argument separates the geometry of the sweep from the additional polynomiality problem required for a Keller lift.

1. Introduction and main conclusions

Consider the polynomial equation

$$(G_y - \mathcal{L}_F F)^2 + 4F_y \mathcal{L}_F G = 0, \quad \mathcal{L}_F = \partial_x + 2F\partial_\theta, \quad (1.1)$$

for $F, G \in k[x, y, \theta]$, where k has characteristic zero. This is a normalization of the middle branch associated with the mixed direction field $(1, x, x^2, y)^\top$ in Gao's construction [1, Section 4.5.2]. We give the normalization explicitly in Section 2.

For any solution define $X : \mathbb{A}_k^3 \rightarrow \mathbb{A}_k^4$ and the sweep $S : \mathbb{A}_k^4 \rightarrow \mathbb{A}_k^4$ by

$$\begin{aligned} X_1 &= -\frac{1}{2}(G_y + \mathcal{L}_F F), & X_2 &= F + xX_1, \\ X_3 &= 2xF - \theta + x^2X_1, & X_4 &= G + yX_1, \\ S(\gamma, x, y, \theta) &= X(x, y, \theta) + \gamma(1, x, x^2, y)^\top. \end{aligned} \quad (1.2)$$

The word sweep in this paper refers to this four-dimensional map, before adjoining any auxiliary coordinate or applying a monomial twist.

Our imposed ansatz is

$$F = (\theta + s(x))^d(y + a(x)) + b(x), \quad a, b, s \in k[x], \quad d \geq 1. \quad (1.3)$$

Theorems 3.1, 4.1, and 5.2 give the following conclusions.

1. A polynomial G exists exactly when $a''' = s''' = 0$, $b = -s'/2$, and k contains a root μ of $Q_d(T) = 9T^2 + 4(d-2)T + 4d^2$. All such G are displayed in (3.4).
2. Every resulting sweep is dominant, has $\det J(S) = \gamma^2$, and has geometric degree $2d + 2$.
3. Over $\mathbb{C}$, its reduced critical image has dimension three. It is therefore not polynomially left–right equivalent to any map

$$S_{\text{susp}}(\gamma, u, v, w) = (T_0(\gamma, u, v), \gamma w + h(u, v)), \quad \det J(T_0) = c\gamma, \quad c \neq 0. \quad (1.4)$$

Here polynomial left–right equivalence means $\tilde{S} = B \circ S \circ A$, with A, B polynomial automorphisms of the source and target. Geometric degree means the degree of the induced function-field extension after extension to an algebraic closure. All critical loci and image closures used below are reduced algebraic sets.

Gao’s Open Problem 4.8 asks for a non-suspension solution of the middle branch, without qualifying the word equivalent there [1]. Our conclusion answers its existence question under the polynomial left–right convention just stated and with the suspension class (1.4). No broader equivalence relation is implicit. In particular, a sweep with Jacobian γ^2 is not a Keller map; the polynomiality conditions required to turn a sweep into a constant-Jacobian map are a separate problem.

This classification differs from the classification by grading signature in Shaska [3] and the graded Keller-map families and quotient-cover invariants of Kistner–Shaska [2]. Those works concern weight-equivariant constant-Jacobian maps. Here the object is a specified polynomial equation for a mixed direction field, and the separating locus is the critical image of the full sweep. Thus neither the use of an equivalence invariant nor the existence of degree families in general is the contribution claimed here.

2. Normalization of the mixed middle branch

We record the coordinate bridge to ensure that the ansatz is attached to the intended branch. Let $c \in k^\times$, let $w = (w_1, w_2, w_3)$, and set

$$\Delta = (1, w_1, w_1^2, w_2)^\top, \quad G_2 = X_2 - w_1X_1, \quad G_3 = X_3 - w_1^2X_1, \quad G_4 = X_4 - w_2X_1.$$

Put $H = G_3 - 2w_1G_2$. The relevant sweep-pencil coefficients are

$$\begin{aligned} D_2 &= -\partial_3 H, \\ D_1 &= (\partial_2 H)(\partial_3 G_4) - (\partial_3 H)(\partial_2 G_4) \\ &\quad - (\partial_1 G_2)(\partial_3 G_3) + (\partial_1 G_3)(\partial_3 G_2), \end{aligned} \quad (2.1)$$

$$D_0 = \det J_{(w_1, w_2, w_3)}(G_2, G_3, G_4). \quad (2.2)$$

The middle-branch conditions are

$$D_2 = c, \quad X_1 = -\frac{D_1}{2c}, \quad 4cD_0 = D_1^2. \quad (2.3)$$

These are the potential form of the branch conditions in [1, Section 4.5]; the following calculation is included in full, so the subsequent classification uses only displayed identities.

Proposition 2.1 (Coordinate bridge and determinant). *A triangular polynomial change of parameters and a nonzero scalar target change transform (2.3) into (1.1). For arbitrary normalized F, G , the map (1.2) satisfies*

$$\det J(S) = \gamma^2 - \frac{1}{4}[(G_y - \mathcal{L}_F F)^2 + 4F_y \mathcal{L}_F G]. \quad (2.4)$$

The variables of the Jacobian are ordered (γ, x, y, θ) .

Proof. The equation $D_2 = c$ gives $H = -cw_3 + \varphi(w_1, w_2)$. Put

$$x = w_1, \quad y = w_2, \quad \theta = w_3 - \varphi(w_1, w_2)/c.$$

The inverse map is $\Psi(x, y, \theta) = (x, y, \theta + \varphi(x, y)/c)$, with determinant one. Temporarily write $F = G_2 \circ \Psi$, $G = G_4 \circ \Psi$, $L = c\partial_x + 2F\partial_\theta$, $A = LF$, and $B = LG$. Then $G_3 \circ \Psi = 2xF - c\theta$, and a row operation gives

$$D_0 = \det \begin{pmatrix} F_x & F_y & F_\theta \\ 2F & 0 & -c \\ G_x & G_y & G_\theta \end{pmatrix} = AG_y - F_y B.$$

For example, the old derivatives of G_2 pull back to $F_x - \varphi_x F_\theta/c$, $F_y - \varphi_y F_\theta/c$, and F_θ . Using the same expressions for G_4 in (2.1), the φ_x, φ_y terms cancel, leaving $D_1 = cG_y + A$. Consequently the last equation of (2.3) is

$$(cG_y - LF)^2 + 4cF_y LG = 0.$$

Now put $f = F/c$, $g = G/c$, and $\mathcal{L}_f = \partial_x + 2f\partial_\theta$. Since $LF = c^2 \mathcal{L}_f f$, $LG = c^2 \mathcal{L}_f g$, $cG_y = c^2 g_y$, and $F_y = c f_y$, the last expression is c^4 times (1.1) for (f, g) . At the level of sweeps this is the left-right change $S^{\text{norm}}(\eta, x, y, \theta) = c^{-1}S(c\eta, \Psi(x, y, \theta))$.

Rename the normalized potentials F, G , and now put $A = \mathcal{L}_F F$, $B = \mathcal{L}_F G$. We have $D_0 = AG_y - F_y B$, $D_1 = G_y + A$, $D_2 = 1$, and $X_1 = -D_1/2$. Expansion of the determinant in (1.2), using its first column $(1, x, x^2, y)^\top$ and the three remaining derivative columns, gives

$$\det J(S) = D_0 + (X_1 + \gamma)D_1 + (X_1 + \gamma)^2.$$

Completing the square proves (2.4). □

3. Classification in the affine-product ansatz

Set $\tau = \theta + s(x)$ and $z = y + a(x)$. Derivatives with respect to x in these coordinates hold z, τ fixed. Thus $\mathcal{L}_F$ becomes

$$\mathcal{D} = \partial_x + a'(x)\partial_z + (s'(x) + 2F)\partial_\tau,$$

and (1.1) becomes

$$(G_z - \mathcal{D}F)^2 + 4F_z \mathcal{D}G = 0. \quad (3.1)$$

Theorem 3.1 (Complete classification within the ansatz). *Let k have characteristic zero, $d \geq 1$, and $a, b, s \in k[x]$. For $F = \tau^d z + b(x)$, a polynomial $G \in k[x, \tau, z]$ satisfies (3.1) if and only if*

$$b = -\frac{1}{2}s', \quad a''' = s''' = 0, \quad (3.2)$$

and, for a root $\mu \in k$ of

$$Q_d(T) = 9T^2 + 4(d-2)T + 4d^2, \quad (3.3)$$

and a constant $\kappa \in k$, it has the form

$$G = \mu \tau^{2d-1} z^3 - \left(a' \tau^d + \frac{s''}{2} \right) z + \frac{a''}{2} \tau + \frac{s'' a}{2} + \kappa. \quad (3.4)$$

For fixed a, b, s, d satisfying (3.2), the solutions are the disjoint union of one affine κ -line for each root of Q_d lying in k .

Proof. Put $h = s' + 2b$ and $m = 2d - 1$. Directly,

$$\mathcal{D}F = 2d\tau^m z^2 + dh\tau^{d-1}z + b' + a'\tau^d. \quad (3.5)$$

If $n = \deg_z G \geq 4$, the leading term of G_z^2 has degree $2n - 2 > n + 1$, whereas all other terms in (3.1) have degree at most $n + 1$. Its nonzero leading square cannot cancel. If $n \leq 2$, including $G = 0$, the nonzero term $4d^2\tau^{4d-2}z^4$ from $(\mathcal{D}F)^2$ cannot cancel. Hence $\deg_z G = 3$.

Write $G = g_3 z^3 + g_2 z^2 + g_1 z + g_0$, with $g_i \in k[x, \tau]$, and put

$$A_2 = 3g_3 - 2d\tau^m, \quad A_1 = 2g_2 - dh\tau^{d-1}, \quad A_0 = g_1 - b' - a'\tau^d.$$

If $\mathcal{D}G = \sum_{i=0}^4 C_i z^i$, then

$$\begin{aligned} C_4 &= 2\tau^d g_{3,\tau}, \\ C_3 &= g_{3,x} + hg_{3,\tau} + 2\tau^d g_{2,\tau}, \\ C_2 &= g_{2,x} + 3a'g_3 + hg_{2,\tau} + 2\tau^d g_{1,\tau}, \\ C_1 &= g_{1,x} + 2a'g_2 + hg_{1,\tau} + 2\tau^d g_{0,\tau}, \\ C_0 &= g_{0,x} + a'g_1 + hg_{0,\tau}. \end{aligned}$$

The five coefficient equations are

$$A_2^2 + 4\tau^d C_4 = 0, \quad (3.6)$$

$$2A_2 A_1 + 4\tau^d C_3 = 0, \quad (3.7)$$

$$A_1^2 + 2A_2 A_0 + 4\tau^d C_2 = 0, \quad (3.8)$$

$$2A_1 A_0 + 4\tau^d C_1 = 0, \quad (3.9)$$

$$A_0^2 + 4\tau^d C_0 = 0. \quad (3.10)$$

The cubic coefficient. Equation (3.6) is

$$(3g_3 - 2d\tau^m)^2 + 8\tau^{2d} g_{3,\tau} = 0. \quad (3.11)$$

If $\deg_\tau g_3 > m$, the square has a unique highest-degree term; if $\deg_\tau g_3 < m$, the term $4d^2\tau^{2m}$ is uniquely highest. Thus its degree is m . Its leading coefficient $\lambda(x)$ satisfies $Q_d(\lambda(x)) = 0$, forcing λ constant by comparison of x -degrees. Call it μ . The discriminant of Q_d is

$$-64(2d-1)(d+1),$$

a negative rational number, so Q_d has no root in $\mathbb{Q}$. If a highest lower term $r_q(x)\tau^q$, $q < m$, remains in $g_3 - \mu\tau^m$, its coefficient in (3.11), at degree $m + q$, is $(6(3\mu - 2d) + 8q)r_q$. The scalar cannot vanish without making μ rational. Consequently $g_3 = \mu\tau^{2d-1}$.

The quadratic coefficient and compensator. Put $\alpha = 3\mu - 2d$ and

$$K = -\frac{(d-2)\mu + 2d^2}{6\mu - 4}.$$

Substituting into (3.7) gives

$$4\tau g_{2,\tau} + 2\alpha g_2 + ((d-2)\mu + 2d^2)h\tau^{d-1} = 0.$$

For each integer $j \geq 0$, $4j + 2\alpha \neq 0$, again because $\mu \notin \mathbb{Q}$. Therefore $g_2 = Kh\tau^{d-1}$. The coefficient of τ^{2d-2} in (3.8) is

$$((2K - d)^2 + 4K(d - 1))h^2.$$

All terms involving g_1, b', h' have degree at least $2d - 1$. Reduction using $Q_d(\mu) = 0$ gives

$$(2K - d)^2 + 4K(d - 1) = \frac{(d+1)^2\mu^2}{(3\mu - 2)^2} \neq 0. \quad (3.12)$$

To verify the reduction without division ambiguity, multiply the difference of the two sides by $(3\mu - 2)^2$: the result is $(d - 1)(d + 1)Q_d(\mu)$. Neither μ nor $3\mu - 2$ vanishes. Thus $h = 0$, so $b = -s'/2$ and $g_2 = 0$.

The remaining coefficients. Equation (3.8) is now

$$4\tau g_{1,\tau} + \alpha g_1 + \frac{\alpha s''}{2} + (3\mu + 2d)a'\tau^d = 0.$$

The nonresonant coefficients $4j + \alpha$ force $g_1 = -a'\tau^d - s''/2$. Equation (3.9) then reads

$$-a''\tau^d - \frac{s'''}{2} + 2\tau^d g_{0,\tau} = 0.$$

Its constant coefficient gives $s''' = 0$, since $d \geq 1$, and hence $g_0 = a''\tau/2 + k(x)$. Finally (3.10) simplifies to

$$2\tau^d(a'''\tau + 2k' - a's'') = 0.$$

It follows that $a''' = 0$ and $k = s''a/2 + \kappa$. This proves necessity and (3.4).

Conversely, under (3.2) and (3.4), differentiation gives

$$\begin{aligned} \mathcal{D}F &= 2d\tau^{2d-1}z^2 + a'\tau^d - s''/2, \\ G_z &= 3\mu\tau^{2d-1}z^2 - a'\tau^d - s''/2, \\ \mathcal{D}G &= (3\mu - 2d)a'\tau^{2d-1}z^2 - (a')^2\tau^d + 2\mu(2d - 1)\tau^{3d-2}z^4. \end{aligned}$$

The left side of (3.1) reduces to $Q_d(\mu)\tau^{4d-2}z^4 = 0$. Distinct roots give distinct cubic coefficients, which also proves the assertion about the solution set. $\square$

Remark 3.2 (Arithmetic scope). The roots are

$$\mu_{\pm} = \frac{2(2 - d) \pm 4\sqrt{-(2d - 1)(d + 1)}}{9}.$$

They are distinct and nonreal. Thus there is no solution in this ansatz over a characteristic-zero field containing neither root, in particular over $\mathbb{Q}$ or $\mathbb{R}$. The classification does not address positive characteristic or $d = 0$, where the degree and nonresonance arguments change.

4. The generic fiber and its degree

Assume the classified data, and write $a = a_0 + a_1x + a_2x^2$, $s = s_0 + s_1x + s_2x^2$.

Theorem 4.1. *The sweep (1.2) is dominant and generically finite of geometric degree $2d + 2$.*

Proof. Proposition 2.1 gives $\det J(S) = \gamma^2 \neq 0$, which in characteristic zero implies dominance. Write R_1, R_2, R_3, R_4 for algebraically independent target coordinates. The first three sweep equations give

$$T(x) = (s_2 - R_1)x^2 + (s_1 + 2R_2)x + s_0 - R_3, \quad (4.1)$$

$$N(x) = R_2 - xR_1 + s'(x)/2 = T'(x)/2, \quad (4.2)$$

$$\theta = 2xR_2 - x^2R_1 - R_3, \quad \tau = T(x). \quad (4.3)$$

On $T \neq 0$ we recover uniquely

$$z = \frac{N}{T^d}, \quad y = z - a(x), \quad \gamma = R_1 - X_1(x, y, \theta). \quad (4.4)$$

The fourth equation, multiplied by T^{d+1} , is

$$P_R(x) = \Lambda_R T^{d+1} + \frac{\mu}{8}(T')^3 - \frac{1}{4}TT'T'' = 0, \quad (4.5)$$

where

$$\Lambda_R = a_2(s_0 - R_3) - \frac{a_1}{2}(s_1 + 2R_2) + a_0(s_2 - R_1) + \kappa - R_4. \quad (4.6)$$

Indeed, before collecting terms the fourth equation is

$$\frac{\mu N^3}{T^{d+1}} - \frac{a'T'}{2} + (R_1 - s_2)\frac{T'}{2T^d} + \frac{a''T}{2} + (s_2 - R_1)a + \kappa - R_4 = 0.$$

Use $N = T'/2$, $R_1 - s_2 = -T''/2$, and the constant identity

$$a''T - a'T' + aT'' = 2a_2(s_0 - R_3) - a_1(s_1 + 2R_2) + 2a_0(s_2 - R_1)$$

to obtain (4.5).

The last two terms of P_R have degree at most three in x . The first has degree $2d+2$, with nonzero leading coefficient: Λ_R contains $-R_4$. Therefore $\deg_x P_R = 2d + 2$.

Let $K_0 = k(R_1, R_2, R_3)$, and write $P_R = -R_4T^{d+1} + Q$, with $Q \in K_0[x]$. The discriminant of T is

$$(s_1 + 2R_2)^2 - 4(s_2 - R_1)(s_0 - R_3) \neq 0,$$

so $\gcd(T, T') = 1$. Since $Q \equiv \mu(T')^3/8 \pmod{T}$ and $\mu \neq 0$, the coefficients T^{d+1} and Q are coprime. Hence P_R is primitive in $K_0[x][R_4]$ and linear in R_4 ; Gauss's lemma makes it irreducible there. It has positive x -degree, so it cannot divide a nonzero member of $K_0[R_4]$. Localizing at the latter elements proves irreducibility in $k(R_1, R_2, R_3, R_4)[x]$.

Let $K = k(R_1, R_2, R_3, R_4)$ and $L = k(\gamma, x, y, \theta)$. The recovery formulas give $L = K(x)$, and the irreducible polynomial P_R gives $[L : K] = 2d + 2$. The same argument applies with k replaced by its algebraic closure: all nonvanishing and coprimality assertions remain valid. This proves the geometric-degree statement.

For the fiber interpretation, a solution with $T = \tau = 0$ must also have $N = T^d z = 0$, impossible generically since $N = T'/2$ and $\gcd(T, T') = 1$. Away from the zero loci of

$$\text{lc}_x(P_R), \quad \text{disc}_x(T), \quad \text{Res}_x(P_R, T), \quad \text{disc}_x(P_R),$$

the polynomial P_R has $2d + 2$ distinct roots with $T \neq 0$, and (4.4) gives exactly one preimage for each root. These excluded polynomials are nonzero by the preceding arguments and separability in characteristic zero. $\square$

5. Critical images and non-suspension

We now work over $\mathbb{C}$. For a polynomial self-map P , set

$$\chi(P) = \dim \overline{P(\text{Crit } P)},$$

where the bar is Zariski closure.

Lemma 5.1. *The integer $\chi(P)$ is invariant under polynomial left–right equivalence.*

Proof. If $\tilde{P} = B \circ P \circ A$, the Jacobian determinants of the automorphisms A, B are nonzero constants. The chain rule gives $\text{Crit } \tilde{P} = A^{-1}(\text{Crit } P)$, so

$$\tilde{P}(\text{Crit } \tilde{P}) = B(P(\text{Crit } P)).$$

The isomorphism B preserves dimensions of closures. $\square$

Theorem 5.2. *Every complex sweep from Theorem 3.1 has $\chi(S) = 3$. Every suspension (1.4) has $\chi(S_{\text{susp}}) \leq 2$. They are not polynomially left–right equivalent.*

Proof. For the classified solution,

$$G_y + \mathcal{L}_F F = (3\mu + 2d)\tau^{2d-1}z^2 - s''.$$

The change $(x, y, \theta) \mapsto (x, z, \tau)$ has determinant one. Differentiating the first three entries of (1.2) gives

$$\det \frac{\partial(X_1, X_2, X_3)}{\partial(x, z, \tau)} = \frac{(3\mu + 2d)(3\mu + 2d - 2)}{2} \tau^{4d-2} z^3. \quad (5.1)$$

One can see the cancellation of the base polynomials directly by using $X_2 - xX_1 = F$ and $X_3 - 2xX_2 + x^2X_1 = -\theta$ in the derivative rows. Modulo $Q_d(\mu) = 0$, the scalar in (5.1) is $(4d + 1)\mu - 2d$, which is nonzero because $\mu \notin \mathbb{Q}$. Thus X has generic rank three and its image closure has dimension three.

The identity $\det J(S) = \gamma^2$ makes the reduced critical locus precisely $\{\gamma = 0\}$; on that locus $S = X$. Therefore $\chi(S) = 3$. For (1.4) the determinant is $c\gamma^2$, and its restriction to the critical locus is $(T_0(0, u, v), h(u, v))$, independent of w . Its image closure has dimension at most two. Lemma 5.1 proves inequivalence. $\square$

Corollary 5.3. *For every even integer $e \geq 4$, there is a polynomial solution of (1.1) whose sweep has geometric degree e and is not polynomially left–right equivalent to (1.4).*

Proof. Take $d = (e - 2)/2$, either complex root of Q_d , and $a = s = \kappa = 0$. Apply Theorems 4.1 and 5.2. $\square$

6. Examples and the boundary of the result

For the simplest choice $a = s = \kappa = 0$, the data become

$$F = \theta^d y, \quad G = \mu \theta^{2d-1} y^3, \quad X_1 = -\frac{3\mu + 2d}{2} \theta^{2d-1} y^2.$$

The other coordinates follow immediately from (1.2). For $d = 1$, the two branches have $\mu = (2 \pm 4\sqrt{-2})/9$ and geometric degree four. For $d = 2$, they have $\mu = \pm 4i/3$ and geometric degree six. Both are non-suspended in the precise sense of Theorem 5.2.

The polynomial P_R describes generic fibers, not every special fiber without further analysis: its leading coefficient or its denominators can degenerate. Likewise, the classification fixes the form (1.3); it does not say every solution of (1.1) can be brought to that form. The equivalence obstruction concerns the four-dimensional sweeps, and does not by itself prove a statement about identity stabilizations of a resulting five-dimensional Keller map. Establishing compatible polynomial twists for every d is not part of these conclusions.

Manuscript information

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References

- [1] S. Gao, *Counterexamples to the Jacobian conjecture in dimensions greater than two*, arXiv:2608.00222v1 (2026), arXiv:2608.00222v1.
- [2] K. Kistner and T. Shaska, *Orbits and fields of definition for graded Keller maps*, arXiv:2608.02863v1 (2026), arXiv:2608.02863v1.
- [3] T. Shaska, *Graded Keller maps and the Jacobian Conjecture*, arXiv:2607.20210v2 (2026), arXiv:2607.20210v2.